

Detailed Report: Oasis Tower RWA Pipeline

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Role: AI Agent & OpenClaw Research Intern

Company: Elchai Group

Date: June 2026

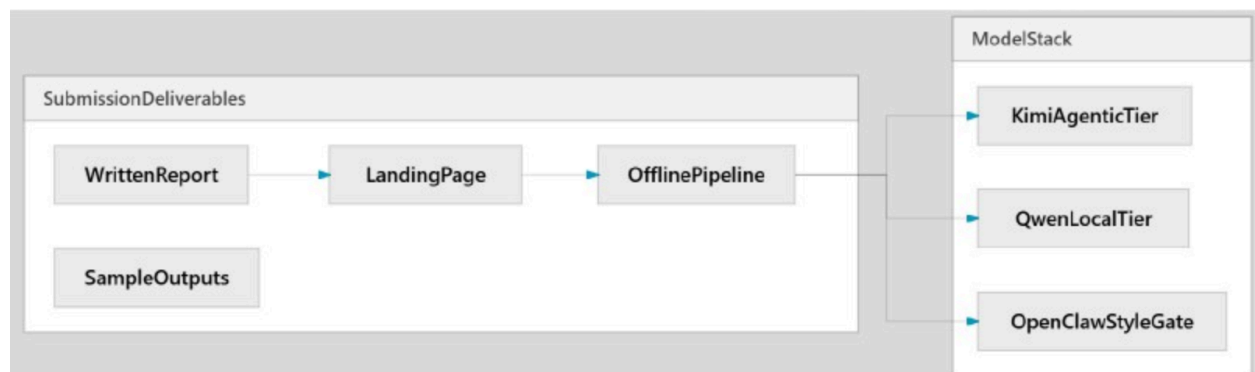
Executive Summary

In this project, the focus is on determining the value of implementing an open source/open weight AI model workflow to assist Elchai Group in a working business setting. In place of the research briefing alone, I created a miniature working Proof of Concept (PoC) for a case study in fractionalizing real estate into tokens: the Oasis Tower RWA Pipeline, a landing page and offline agent workflow and browser-based demo.

The selected model/Tool is an agentic reasoning level provided by Kimi which runs together with an orchestration pattern from OpenClaw and a local Qwen/Ollama compatible sovereign level. In the default project, it does not require any API key and has deterministic mock logic, local RAG and regex redaction enabled so that the reviewer can test the entire project without model credentials.

Recommendation: Internal research Limited Proof of Concept (limited demo) with guardrails that are available for synthetic applications, client workshops and internal pilots

RWA discovery demos. Use of real client data will not be permitted until Elchai has finished legal, infrastructure, security and model provider review of the data.



1. Selected Model / Tool

Primary Selection: Kimi

Kimi was chosen because the reference video focused on Chinese open-source / open-weight model progress and specifically mentioned Kimi as strong for **agentic, multi-step work**.

Task type	Kimi role in this POC
Compliance mapping	Map asset facts to review-required topics
Architecture drafting	ERC-3643-style permissioned token outline
Multi-stage reasoning	Chain outputs across pipeline stages
Structured output	JSON manifests for human review

Supporting Tool: OpenClaw-style orchestration

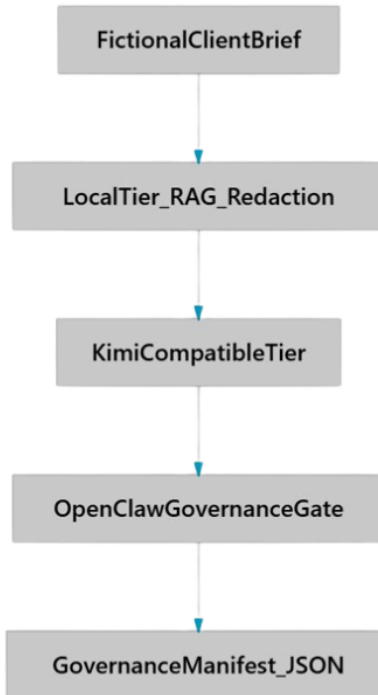
OpenClaw matches the job title and the business need: staged agents, source grounding, model routing, and human control, not a single chat window.

Local Tier: Qwen / Ollama-ready (with free fallback)

Paid tools are not required for the assessment.

The default demo uses:

- Local RAG over fictional corpus files ([agent/rag_corpus/] (agent/rag_corpus/))
- Regex redaction ([agent/local_qwen.py] (agent/local_qwen.py))
- Deterministic mock outputs ([agent/rwa_mock.py] (agent/rwa_mock.py), [web/engine.js] (web/engine.js))
- Optional live Kimi path ([agent/kimi_client.py] (agent/kimi_client.py))



Tier	Intended tool	Default demo behavior
Sovereign	Qwen via Ollama	Local RAG + regex redaction (offline)
Agentic	Kimi K2-family via API	Deterministic mock (live path optional)
Orchestration	OpenClaw gateway	Python POC + config in openclaw/
Gate	Deterministic halt	Human sign-off before manifest

2. Simple Explanation Of The Tool

In plain terms:

- Kimi-compatible model: Compliance flags, architecture drafts, structured JSON are examples of a Kimi-compatible model supports complex reasoning after data preparation.
- Local RAG / Qwen-ready tier: Performs sensitive document work first: extracts facts, redactions, PII, escalating if not supported.
- OpenClaw-oriented workflow: Stages the AI system with a human approval check gate in front of risky work.

For Elchai: Publicly active in the areas of AI agents, RAG, Smart contracts, Tokenization, Finance, and Real estate. It is not the model; it is the process and procedure of controlling the model.

3. Practical Project Output

Project Name

Oasis Tower RWA Pipeline

Fictional Client

Oasis Tower SPV: Dubai commercial asset considered for fractional tokenization.

Workflow Outputs

Output	Description
Source-grounded asset facts	Extracted from offering memo / brief with citations
Review-required compliance topics	VARA / ADGM / PDPL-style flags (not legal advice)
Permissioned token architecture outline	ERC-3643-style modules, no production Solidity
Escalation questions	Items missing from source documents
Governance manifest	Structured JSON for human review
Human sign-off gate	Pipeline HALTED until authorized approval

Why This Is Practical

Before building a full tokenization platform, a team needs to know: what the asset is, what documents support the facts, what compliance questions exist, what architecture may fit, and what humans must approve. This POC organizes the **first structured draft**, it does not replace lawyers, engineers, or auditors.

Run commands:

Browser

serve_web.cmd → http://localhost:8123

Terminal

cd agent

python orchestrator.py --brief sample_briefs/oasis_tower_rwa.json --yes

4. Landing Page Output

Location: web/index.html

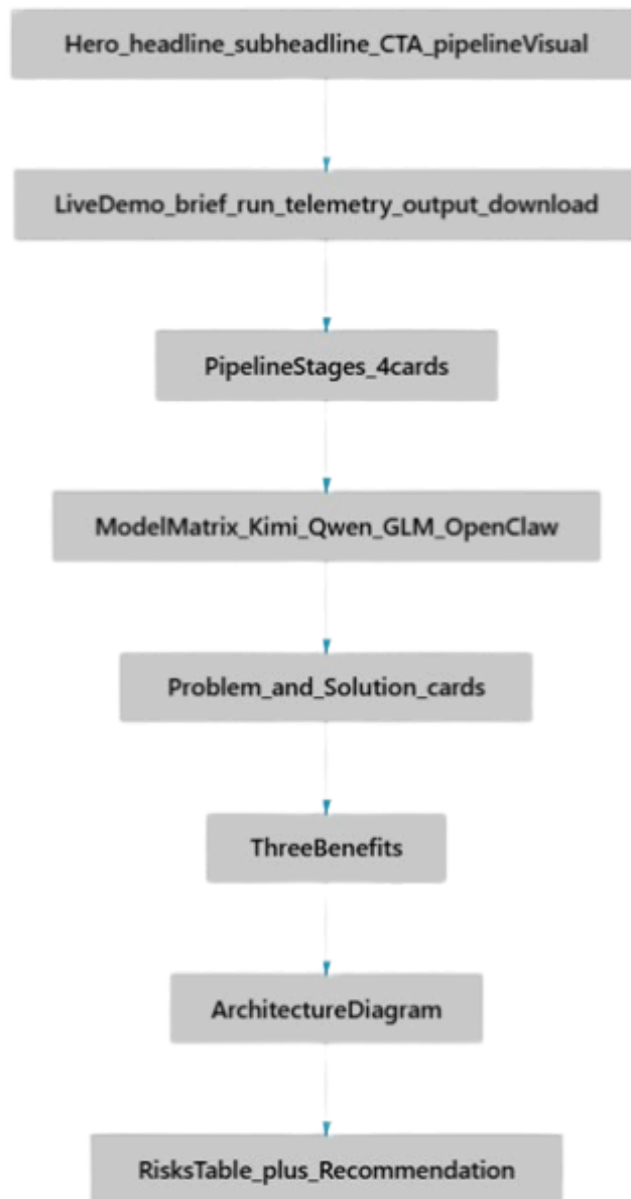
Required Elements (Assessment Checklist)

Required item	Included content
Headline	"Oasis Tower: a RAG-driven fractional real estate tokenization pipeline."
Subheadline	Local sovereign tier + Kimi-compatible tier + OpenClaw-style human gate
Problem	Regulated RWA needs source-grounded facts, privacy, compliance, human approval
Solution	Four-stage staged AI workflow with governance gate
Three benefits	Grounded discovery, privacy-first routing, audit-ready handoff
Use case	Fractional real estate tokenization fictional Oasis Tower SPV
Call to action	"Run RWA pipeline"
Layout / wireframe	Demo-first page structure (see diagram below)

Three Benefits

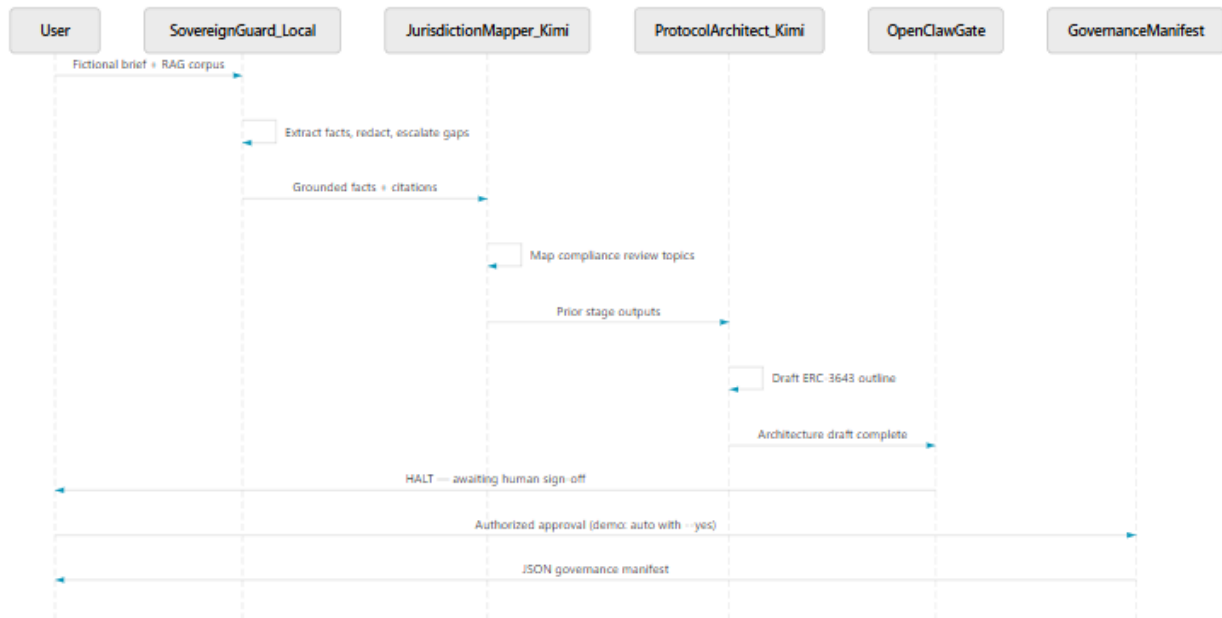
#	Benefit	Meaning
01	Grounded discovery	Source documents and citations instead of free-form guessing
02	Privacy-first routing	Sensitive handling locally before optional external model calls
03	Audit-ready handoff	Structured governance manifest for human review

Page Layout / Wireframe



5. Workflow And Architecture

Four-Stage Pipeline



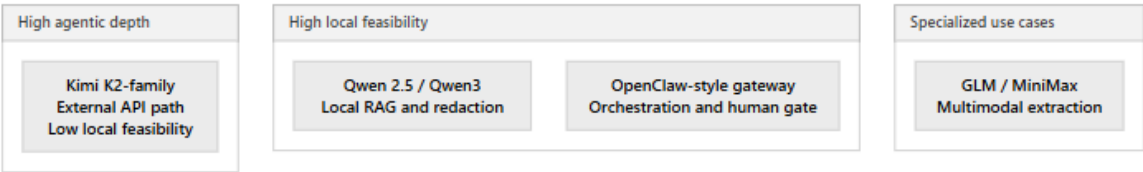
Stage Breakdown

Stage	Agent	Tier	Purpose
0	Local prep	Qwen-ready / offline	RAG retrieve, regex redact, offline reproducibility
1	SovereignGuard	Local	Asset facts, distribution rules, citations, escalations
2	JurisdictionMapper	Kimi-compatible	Review-required compliance flags (not legal advice)
3	ProtocolArchitect	Kimi-compatible	Permissioned token architecture outline (no Solidity)
4	GovernanceManifest	Deterministic	HALTED manifest + next steps

Control pattern: 80% automated draft / 20% human control

6. Model / Tool Evaluation Matrix

Model / Tool	What it is	Best use for Elchai	Local deployment	Main risk
Kimi K2-family	Agentic reasoning model	Multi-step reasoning, architecture drafting, coding	External API in live path	Data residency, vendor approval, reliability
Qwen 2.5 / Qwen3	Open model family	Local RAG, parsing, redaction, sovereign pre-filter	Practical via Ollama (hardware dependent)	Lower ceiling vs large external models on long agentic tasks
GLM / MiniMax	Alternative Chinese model families	Multimodal / specialized extraction (KYC docs, floor plans)	Depends on quantization	Not directly tested in this POC
OpenClaw-style gateway	Agent orchestration pattern	Human-in-the-loop workflows across channels	Local/team service concept	Security hardening and operational ownership required



Selection Rationale

- **Kimi:** Best match for reference video's agentic, long-horizon emphasis; live path kept available.
- **Qwen:** Sovereignty option: local work before external models see client data.
- **OpenClaw-style orchestration:** Stage separation, role prompts, escalation rules, human approval, model routing.

Recommended production pattern:

Local sensitive-data tier + external agentic tier + deterministic human gate

7. Use Case For Elchai Group

Elchai Group is a public presentation of work in the field of AI agents, RAG, enterprise software, smart contracts, tokenization, finance and real estate. This project is centered around a fictional client-facing RWA workflow, a problem which IS NOT invented in Elchai but rather with the customer.

For Oasis Tower, the pipeline:

- Finds information in a local offering memorandum corpus
- Filters out clearly sensitive areas prior to subsequent model use
- Maps asset facts to see review required VARA/ADGM/PDPL reference chunks
- Writes out an outline of ERC-3643 "permissioned" token architecture
- The workflow stops before production blockchain work begins by requiring an OpenClaw-style governance gate and authorized human sign-off.

Could be used as: research POC, client workshop asset, internal engineering template for regulated RWA discovery.

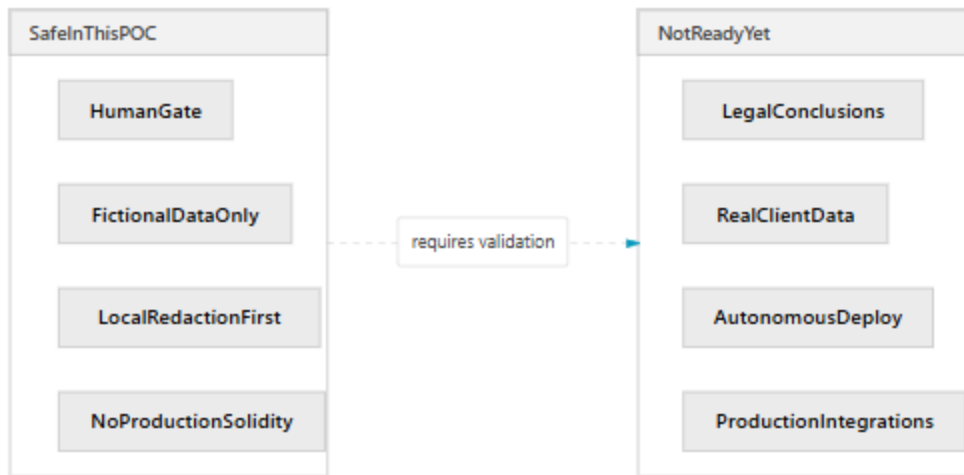
8. Departments That May Benefit

Department	How it benefits
Engineering	Reusable RAG + staged agent workflow for tokenization discovery
IT / Infrastructure	Evaluate local redaction, provider boundaries, deployment controls, security
Product	Working demo explaining abstract RWA workflow to clients/stakeholders
Finance	Compare API cost, local infrastructure cost, open-model tradeoffs
Strategy	Evaluate whether Kimi/Qwen/OpenClaw-style workflows fit GCC enterprise delivery
Operations	Repeatable intake, escalation, and handoff for regulated project discovery
Legal / Compliance	Review-required flags with citations instead of unsupported AI claims

9. Risks And Limitations

The main value is showing **judgment** about when the tool should and should not be used.

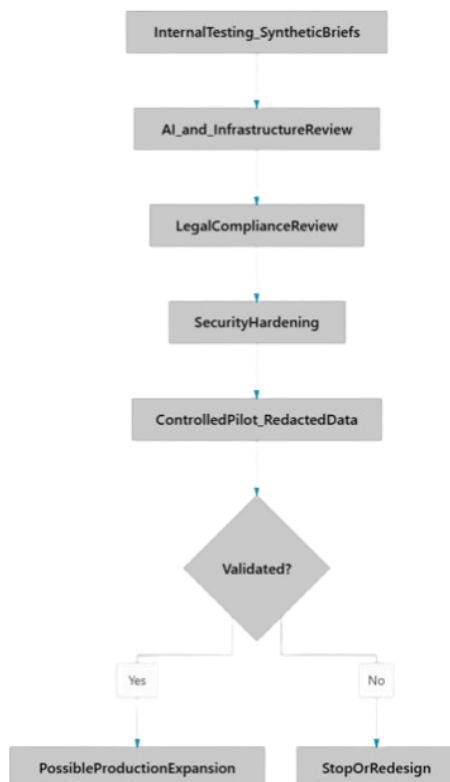
Risk	Why it matters	Mitigation in POC	Remaining limitation
Privacy	Client/investor/KYC data is sensitive	Fictional data only; local redaction first	Real data needs legal + IT approval
Security	Agents can become risky if connected to deployment actions	Human gate; no autonomous deploy; no production Solidity	Real OpenClaw needs hardening
Model accuracy	Wrong summaries mislead teams	Structured prompts, citations, review-required labels	Expert review mandatory
Hallucination	AI invents unsupported facts	Escalate-if-not-in-source rule	Corpus quality limits output
Local hardware cost	Local models need compute	Lightweight offline demo; Qwen optional	Production local inference may need investment
Integration difficulty	Real RWA needs KYC, custody, wallets, audit	POC isolates discovery/architecture stage	Production integrations are separate
Reliability	APIs and parsers fail	Offline mock + deterministic manifest	Live mode depends on provider uptime
Vendor/procurement	External providers may not meet policy	Model-agnostic architecture	Procurement must approve providers



10. Final Recommendation

Recommendation

Test as a limited internal research proof of concept with guardrails.
This does not mean avoid it. It means use it in the right order:



11. Prompts And Tools Used

Tools

Tool / File	Purpose
agent/orchestrator.py	Staged pipeline CLI
web/index.html	Landing page + browser demo
web/engine.js	Browser-side mock pipeline
agent/rwa_mock.py	Deterministic offline logic
agent/brief_parser.py	Custom brief field extraction
agent/rag_store.py	Simple local RAG retrieval
agent/local_qwen.py	Optional Qwen/Ollama helper
agent/kimi_client.py	Optional Kimi live API wrapper
openclaw/	OpenClaw-style deployment concept

Prompt Files

Prompt	Purpose
agent/prompts/sovereign_guard.txt	Local source-grounded extraction and escalation
agent/prompts/jurisdiction_mapper.txt	Compliance review mapping
agent/prompts/protocol_architect.txt	Permissioned token architecture outline

12. Closing Statement

The Oasis Tower RWA Pipeline is not a production replacement for lawyers, engineers, or auditors. It is a safe adoption test. It shows how Elchai could evaluate open-source and open-weight AI models in a realistic business workflow while respecting privacy, security, model limitations, and human governance.

The project does not only show that AI can generate text. It shows how AI can be placed inside a controlled workflow that a real company could evaluate before adoption.

13. References

1. Elchai Group public website: <http://www.elchaigroup.com/>
2. Kimi K2 technical paper: <https://arxiv.org/abs/2507.20534>
3. Qwen2.5 technical report: <https://arxiv.org/abs/2412.15115>
4. Qwen3 technical report: <https://arxiv.org/abs/2505.09388>
5. OpenClaw project/docs: <https://github.com/openclaw/openclaw> and <https://docs.openclaw.ai>